

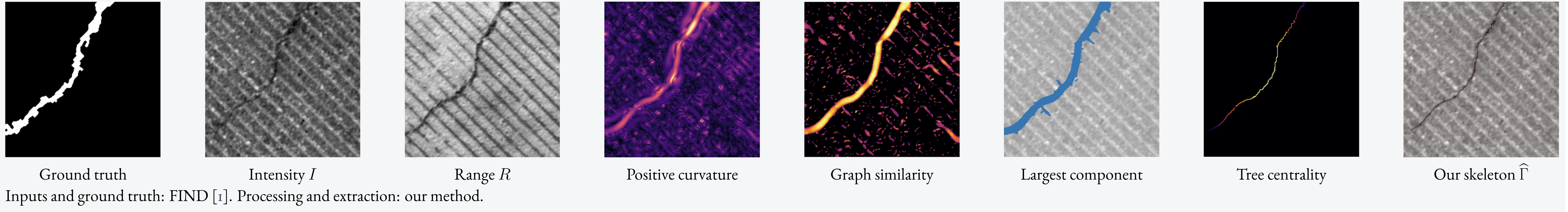
Multi-Modal, Training-Free Crack Extraction via Generalized Frangi Graph

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FIND first example.

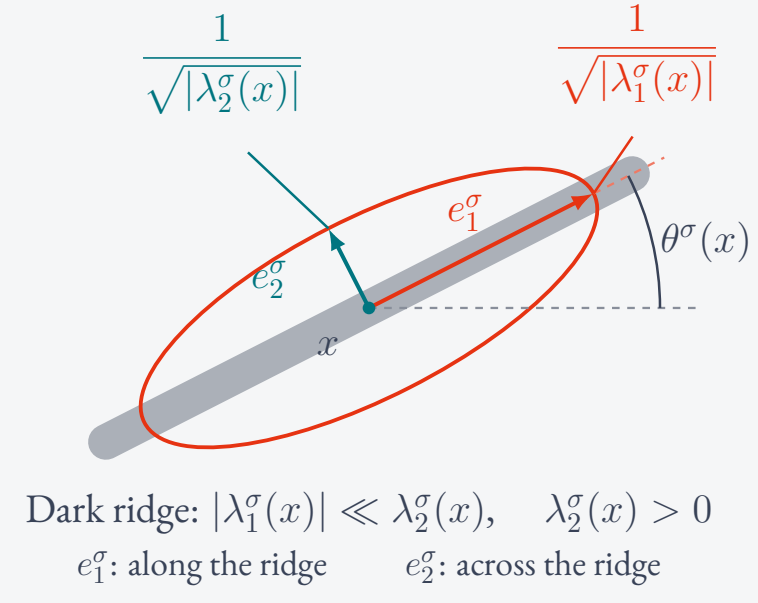


1. Fuse the modalities

Normalize each modality.

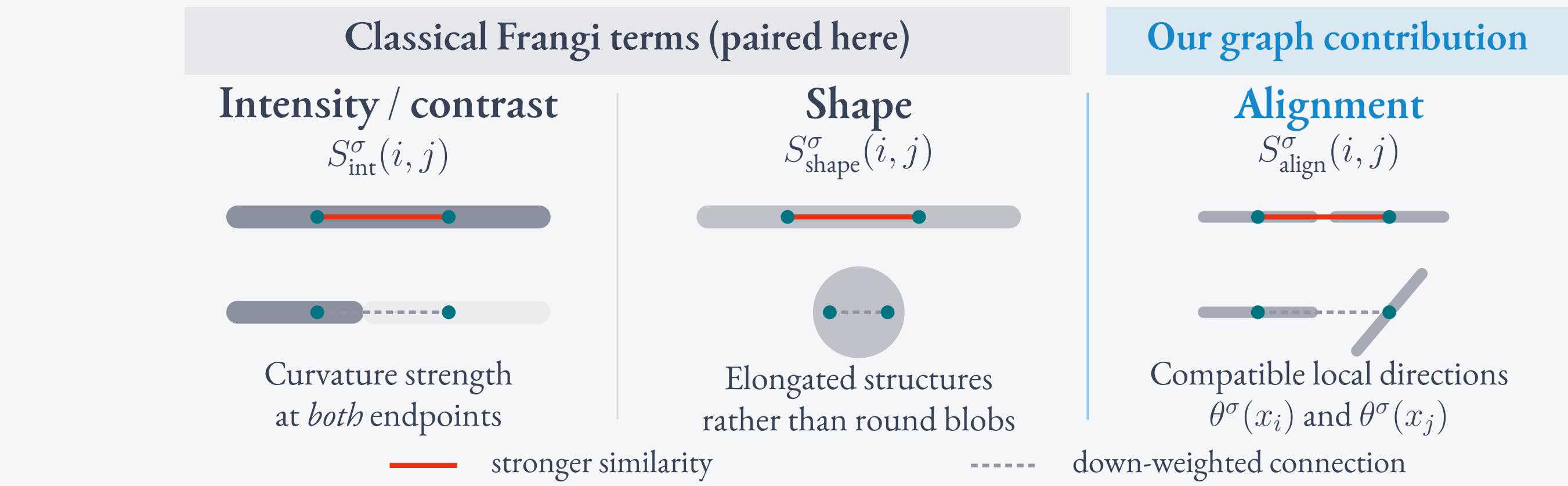
$$\hat{\mathcal{H}}_{\sigma}^{(m)}(x) = \frac{\mathcal{H}_{\sigma}^{(m)}(x)}{\max_{y \in \Omega} \left\| \mathcal{H}_{\sigma}^{(m)}(y) \right\|_2}.$$

Aligned dark-ridge inputs; $w_m \geq 0$, $\sum_m w_m = 1$.



Extract the geometry after fusion.
At scale σ : $|\lambda_1^{\sigma}| \leq |\lambda_2^{\sigma}|$.
Ridge direction: θ^{σ} .

2. From pixel-wise FRANGI to pairwise links



FRANGI [2]: intensity and shape. **Our graph adds alignment.** For $0 < \rho_{ij} = \|x_i - x_j\|_2 \leq R$:

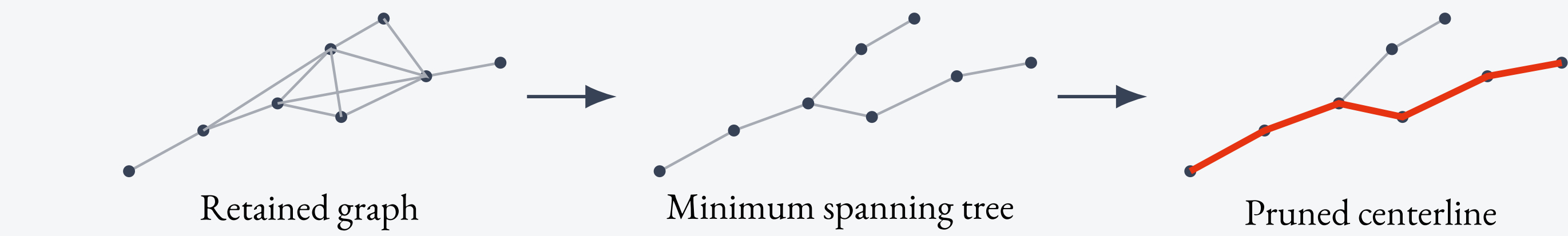
$$C_{\sigma}(x) = \max\{\lambda_2^{\sigma}(x), 0\}, \quad R_B^{\sigma}(x) = \frac{|\lambda_1^{\sigma}(x)|}{|\lambda_2^{\sigma}(x)|}.$$

$$S_{ij}^{(0)} = \max_{\sigma \in \Sigma} S_{\text{int}}^{\sigma}(i, j) S_{\text{shape}}^{\sigma}(i, j) S_{\text{align}}^{\sigma}(i, j)$$

$$d_{ij} = \min\{1, \rho_{ij}(1 - S_{ij}^{(0)})\}, \quad S_{ij} = 1 - d_{ij}.$$

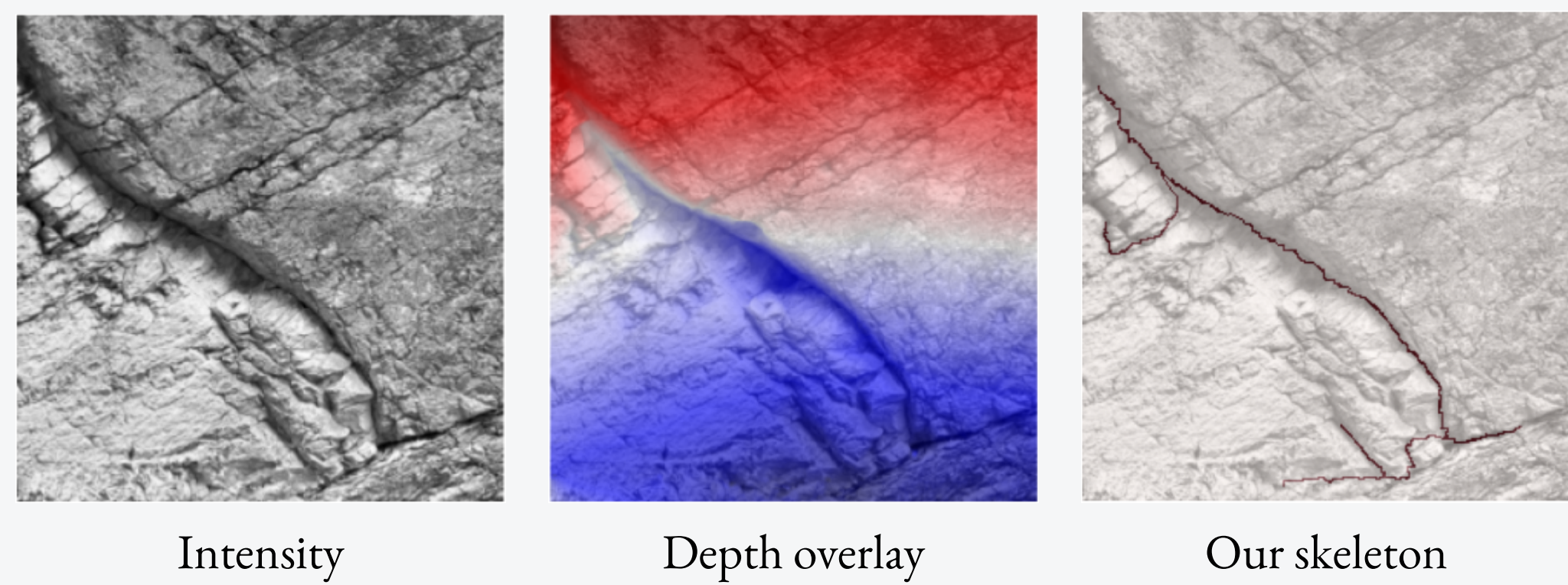
ρ_{ij} penalizes long links; θ and $\theta + \pi$ denote the same axis.

3. Extract the centerline



Per component: minimum spanning tree with costs d_{ij} , then similarity-weighted centrality. FIND settings: $\Sigma = \{1, 3, 5, 7\}$, $R = 3$, retained fractions $\tau_E = \tau_V = 0.25$.

Multimodality: Palais des Papes (France)



Fuse visible intensity and photogrammetric depth.

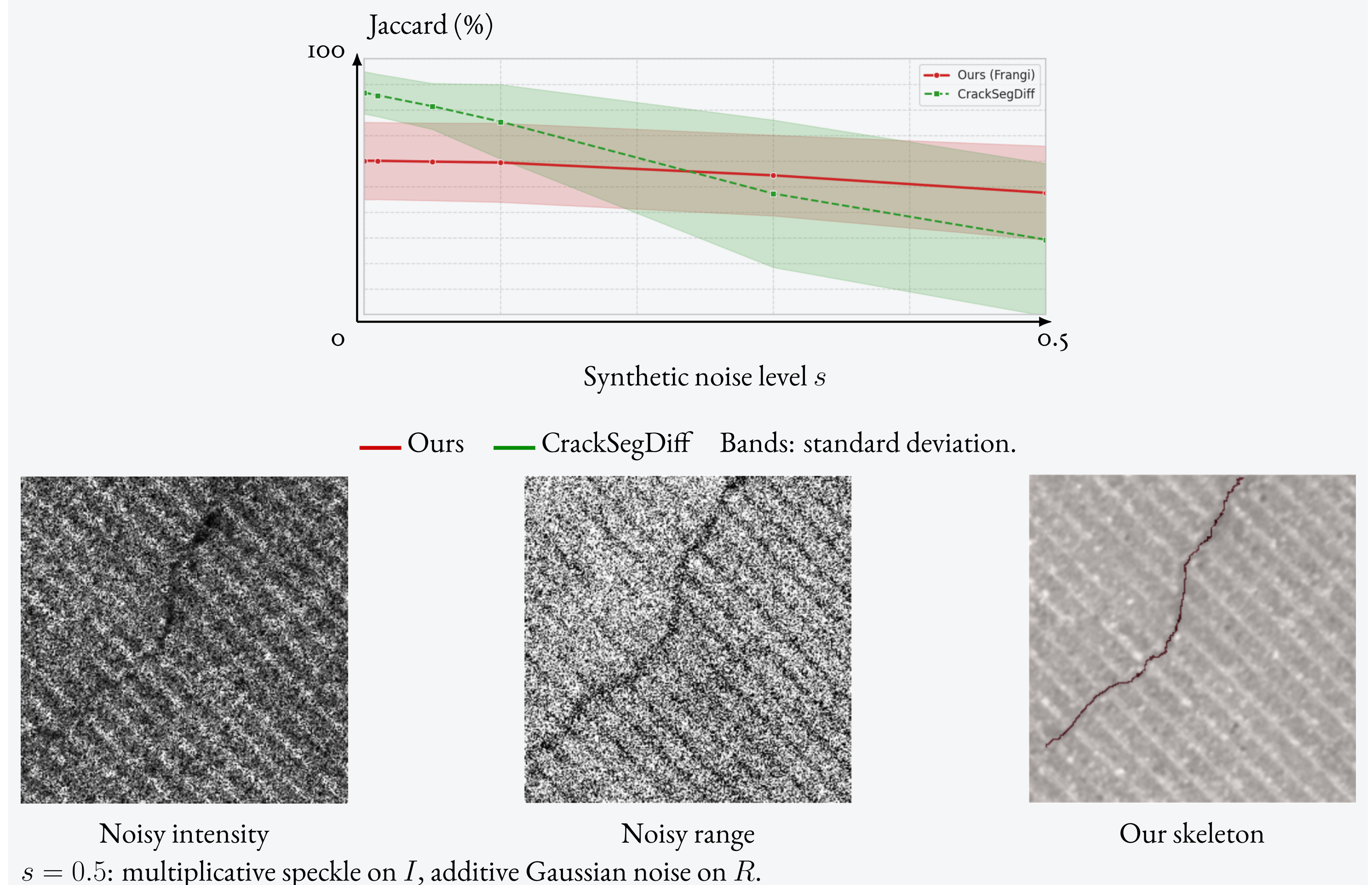
4. FIND: clean data

Method	Jaccard $\uparrow$	Tversky $\uparrow$	Wasserstein $\downarrow$
Ours, no training	63%	71%	11 px
CrackSegDiff [3]	87%	90%	5 px

FIND: 256×256 . Overlap: both skeletons dilated by 3 px; Tversky: $\alpha = 1$, $\beta = 0.5$; Wasserstein: skeleton distance.

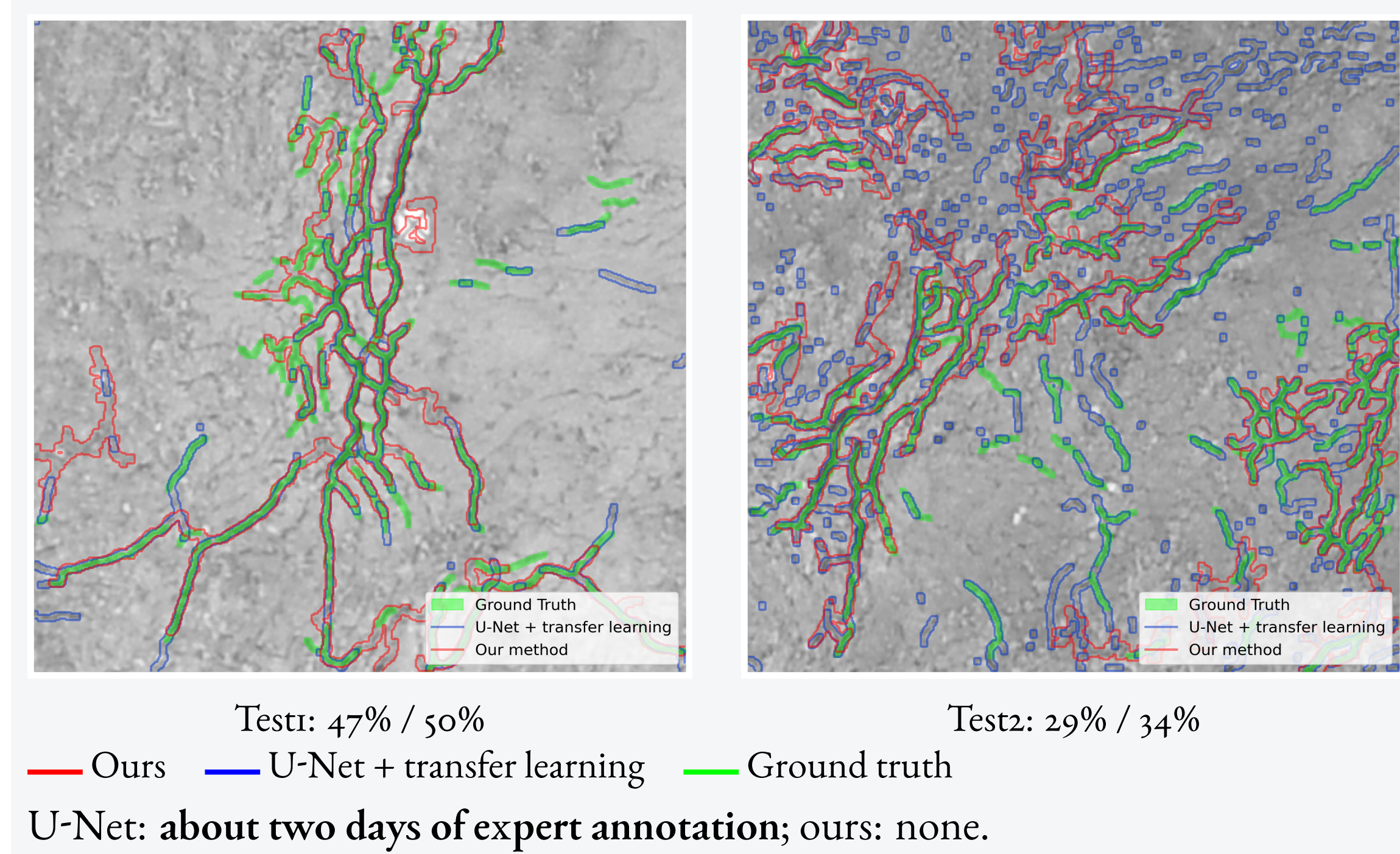
5. FIND: synthetic noise

Slower degradation under the tested synthetic noise.



6. Geological domain shift

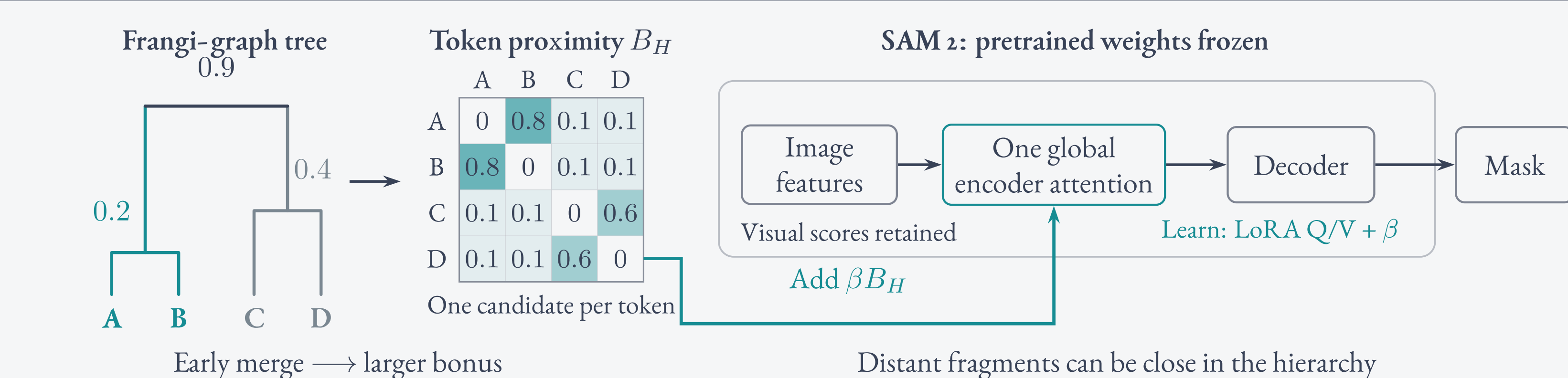
Our evaluation: Vaches Noires. Two Cerema UAV patches (512×512).



Jaccard:
ours / U-Net.
U-Net adapted by us
on a separate image;
transfer inspired by
Shi et al. [4].

U-Net: **about two days of expert annotation**; ours: none.

Perspective: frozen SAM 2 + LoRA + Frangi hierarchy



SAM 2 [5]: frozen pretrained weights.

Train LoRA [6] and our coefficient β together.

$$A_H = \text{softmax}(L + \beta B_H).$$

L : visual scores. **Our proposal:** B_H from Frangi-graph merge heights.

Ideas used: maximin connectivity (MALIS [7]) and attention biases (Graphormer [8]).

CrackSAM [9]: SAM adaptation to cracks.

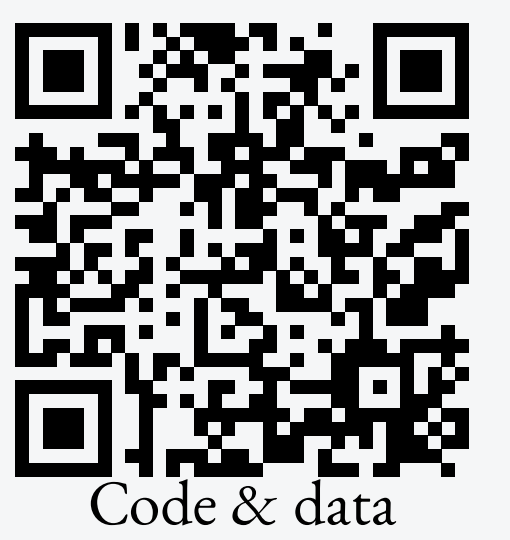
Aim: fewer breaks without more false links.

References

Background review: Zhang et al. [10].

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Code & data